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ORIGINAL ARTICLE

Isometric-based EMG threshold in girls and women

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Abstract

Background The electromyographic threshold (EMG_{Th}) has been suggested to indicate the onset of accelerated higher-threshold (type-II) MU recruitment. Previous research has demonstrated that boys' EMG_{Th} occurs at higher relative exercise intensities than men's in both cycling- and isometric-based testing. Girls−women EMG_{Th} differences were demonstrated only in cycling, but findings were clouded by low EMG_{Th}-detection rates in women (68%) and particularly in girls (45%)

Purpose To examine the EMG_{Th}, in girls and women, using the same males-employed isometric-based test protocol, and compare the females' findings with those previously obtained in the males.

Methods Seventeen girls and 17 women had their EMG_{Th} determined as well as their one-repetition-maximum isometric knee-extension strength (1RM). Vastus-lateralis sEMG root mean square was recorded and the EMG_{Th} was defined as the exercise intensity (%1RM) at the bi-segmental point of the least sum of squares.

Results EMG_{Th} was detected in 88.2% of girls and 94.1% of women and occurred at higher relative intensities in the girls than in women (56.0 ± 11.1 vs. 47.7 ± 8.0% 1RM). The girls' 1RM (normalized to lean body mass) was only 69.1% that of the women.

Conclusions Girls' EMG_{Th} values are higher compared with women's, possibly reflecting lower ability to activate higher-threshold (type-II) motor units. The females' EMG_{Th} and detection-rate values were similar to the corresponding values previously observed in males. The females' age-related difference in the recruitment of higher-threshold motor units, as reflected by the EMG_{Th}, appears to be on par with the males.

Keywords Child · Exercise · Fatigue · Force · Electromyography · Maturation · Muscle · Neuromotor · Torque

Abbreviation

EMG	Electromyography
EMG_{Th}	Electromyographic threshold
EMG_{RMS}	Root mean square of the EMG signal
$EMG_{RMS}5$	Average of EMG_{RMS} values for each load
IICT	Intermittent isometric-contraction test
LBM	Lean body mass
LSS	Least sum of squares
MU	Motor unit
MVC	Maximal volitional contraction
RM	Repetition maximum
RMS	Root mean square
sEMG	Surface EMG

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Introduction

Volitional motor unit (MU) activation has been demonstrated to be lower in children compared with adults (O'Brien et al. 2010). That is, children have a greater MU activation deficit (Grosset et al. 2008). In accordance with known recruitment hierarchy and the Henneman size principle (Henneman et al. 1965; Vollestad et al. 1984), it has been hypothesized that this activation deficit mainly comprises higher-threshold (type-II) MUs (Dotan et al. 2012). The electromyographic threshold (EMG_{Th}) is a commonly used measure, widely suggested to indicate the onset of accelerated higher-threshold (type-II) MU recruitment (e.g., Candotti et al. 2008; Hug

et al. 2003, 2006; Lucia et al. 1999; Maestu et al. 2006). As such, the EMG_{Th} can be taken to reflect the relative utilization of type-II MUs out of the available functional MU pool. The EMG_{Th} has been studied mainly in adult males and primarily in cycling (Camic et al. 2011; Camic et al. 2010; Candotti et al. 2008; Hug et al. 2004; Hug et al. 2006; Hug et al. 2003; Jurimae et al. 2007; Lucia et al. 1999; Maestu et al. 2006; Moritani et al. 1993; Taylor and Bronks 1996). Two recent studies demonstrated that, compared with men, boys' EMG_{Th} occurs at higher relative exercise intensities both in progressive cycling (Pitt et al. 2015), as well as in progressive isometric contractions (Woods et al. 2019). These findings support the hypothesis that children utilize their type-II MU's to a lesser extent compared with adults (Dotan et al. 2012).

Prior to puberty, girls and boys exhibit similar physiological responses to exercise (e.g., maximal strength, muscle endurance, aerobic capacity; (Blimkie et al. 1988; Grosset et al. 2008; Lillegard et al. 1997; O'Brien et al. 2010; Seger and Thorstensson 2000). During maturation, males gain considerably greater muscle mass compared with females (Malina 1969) and there is generally a greater maturational increase in performance capacity in than in females (Blimkie et al. 1988; Falk et al. 2009a, b). Our previous, cycling-based studies (Long et al. 2017; Pitt et al. 2015), indeed suggested a somewhat greater child−adult EMG_{Th} difference in males than in females. However, that comparison and conclusions thereof were hampered by relatively low EMG_{Th} detection rates. While the men's EMG_{Th}-detection rate was 95%, it was only 78% in the boys and much lower in the women (68%) and particularly in the girls (45%).

We have recently developed an intermittent isometric contraction test (IICT), that enhanced boys' EMG_{Th}-detection rates, by > 13% (Woods et al. 2019), presumably by facilitating attainment of greater contractile forces than could be attained by the previously employed cycling protocol (Long et al. 2017; Pitt et al. 2015). Moreover, the isometric contraction protocol also resulted in a greater boys−men EMG_{Th} difference of 10.4%, compared with the 6.6% cycling-based difference. The enhanced detection-rate was suggested to be due to the IICT's ability to detect higher-intensity thresholds, that were undetectable by the cycling-based protocol (i.e., primarily in children).

Thus, the aim of the present study was to utilize the newly developed, more sensitive IICT protocol (Woods et al. 2019), to further investigate higher-threshold MU recruitment (as reflected by the EMG_{Th}) in girls vs. women. Our secondary aim was to compare the age-related EMG_{Th} difference in females with the corresponding, previously published difference in males (Woods et al. 2019). Note that the two studies were conducted in the same laboratory, with identical protocols, and by the same researchers. It was hypothesized that the relative contractile intensities at the EMG_{Th} would be higher in the girls than in the women, that EMG_{Th}-detection rates will be on par with those found in the males, and that the child−adult EMG_{Th} difference would be smaller in the females compared with the males.

Methods

Participants

Healthy, recreationally active girls ($n = 20$, 8−12 years) and women ($n = 18$, 18−35 years) volunteered for the study. The girls were all of maturational stages 1−3 [self-assessment; (Tanner 1962)], and before the age of peak height velocity (Mirwald et al. 2002). Data from three girls were not analyzed due to poor compliance with the test protocol and data of one woman were not used due to corrupted EMG data files, leaving 17 girls and 17 women for the final analysis. Participant characteristics are presented in Table 1.

The study was cleared by Brock University's Research Ethics Board. Prior to signing informed consent and assent forms, a thorough explanation of the study's purpose, measurement procedures, benefits and potential risks or discomforts were provided to all participants and the children's parents/guardians.

Experimental procedure

All tests and procedures were completed at the Applied Physiology Laboratory at Brock University. Anthropometric variables (height, weight, body composition; see "Measurements", below), training history and pubertal stage (girls only) were assessed. Participants were then fitted to the dynamometer and performed maximal strength testing (1RM), followed by EMG_{Th} determination via the IICT protocol, performed to volitional exhaustion. To optimize

Table 1 Participant characteristics

	Girls	Women
n	17	17
Age (years)	10.0 ± 1.6	23.8 ± 2.3****
Body mass (kg)	35.5 ± 12.3	66.3 ± 9.1****
Height (cm)	139.1 ± 11.6	163.9 ± 6.0****
%Fat	17.4 ± 9.3	29.1 ± 7.1****
Pubertal stage# (1,2,3,4,5)	11, 2, 1, 3, 0	NA
Maturity offset (years)	− 3.1 ± 1.4	NA
Activity score	89.0 ± 40.5	63.8 ± 26.0*
Sport training (hrs wk^{-1})	7.6 ± 6.1	6.1 ± 5.0

Values are means ± 1SD

Significance of group difference: **p < 0.05; ****$p < 0.001$

#Determined by secondary sex characteristics (Tanner 1962); $n = 17$

attainment of true maximal torque values, two additional 1RM test sessions were conducted on subsequent days.

Dynamometer

Commercially available isokinetic dynamometers (e.g., Biodex, Cybex) typically have isometric settings, allowing for contractions at fixed joint angles and muscle lengths, but not fixed torque levels. The resultant torques, applied by any given testee, may, therefore, appreciably fluctuate, particularly with children. To obtain more stable, pre-determined torques, a custom dynamometer was built, equipped with a knee-extension lever, connected to a weight rack via a pulley system. Weights could be added or removed to produce discrete torque settings at a 1.02 Nm (0.25 lb) resolution. For each contraction, the participant extended her knee and lifted the weight rack from ~ 110−120° starting position to the final 125−130° position and then held the contraction isometrically until signaled to bring the weights down for an overall total of 5 s. The dynamometer's torque output was calibrated between 16 and 183 Nm against known weights to determine its weight-to-torque conversion factor. The correlation coefficient throughout that range was 0.999. That dynamometer was used for both the 1RM and EMG_{Th} tests.

Measurements

Anthropometric measures included standing and sitting (children only) height to the nearest 0.1 cm (Ellard Instrumentation Ltd. Stadiometer), body mass and fat percentage measured to the nearest 0.1 kg and 0.1%, respectively, by a bioelectrical impedance analysis apparatus (InBody 520, Biospace CO., Ltd., S. Korea). Participants were hydrated upon arrival to the laboratory and asked to void their bladder prior to measurements. Following instruction, pubertal status was self-assessed by the girls, based on secondary sex characteristics (i.e., pubic hair) as outlined by Tanner (Tanner 1962).

Habitual physical activity was assessed using the Godin-Shephard Leisure-Time Exercise Questionnaire (Godin and Shephard 1985). Current and past history of sports and resistance training was self-reported via a questionnaire.

Exercise protocols

All tests were conducted on the right leg. 1RM was determined, as the knee-extension torque at lifting the highest attained weight load out of 2–8 trials with ~ 1-min rest intervals. The 1RM was used to determine the starting load (~ 25% 1RM) and loading progression (~ 3% 1RM) for the IICT protocol. The highest 1RM value of the three visits was used for determining the EMG_{Th} test load increments, as well as EMG_{Th} intensity (%1RM). If the highest 1RM occurred on the second or third visit, then the %1RM of the load increments and EMG_{Th} were retroactively calculated.

The IICT protocol has been previously described (Woods et al. 2019). Briefly, starting at 25% 1RM, participants performed sets of five 5-s isometric contractions, 3-s apart. A 30-s rest was provided between successive sets, incremented by ~ 3% 1RM. The test was terminated upon volitional exhaustion (inability or unwillingness to raise the weight basket off its base in any one or more of the load's five contractions). Since loading was attained by discrete weights, some of the smallest/weakest girls' load increments had to be set closer to 4% to avoid excessively long tests (more stages). For this reason, the mean load incrementation was 3.16 ± 0.61% 1RM for the girls and 2.94 ± 0.18% 1RM for the women; $p > 0.05$). For girls whose 1RM was < 60 Nm, the starting load was ~ 30% 1RM since minimal loading was limited by the weight of the loading rack (1725 g/18.0 Nm).

Data acquisition

EMG and torque data acquisition and reduction have been previously described (Woods et al. 2019). Briefly, data were sampled and recorded at 1000 Hz, (EMGworks Acquisition, Delsys Inc., Boston, MA, USA) and band-pass filtered at 20−450 Hz, using the Bagnoli-4 bioamplifier (Delsys Inc., Boston, MA, USA). The muscle- and reference-electrodes were placed according to SENIAM guidelines for the vastus lateralis. An electro-goniometer (S700/S720 Joint Angle Shape Sensor, Measurand Inc., Fredericton, NB, Canada), attached to the dynamometer across the lever's pivot axis, was used to record knee angle and the minor changes thereof during the holding/isometric phase of the contraction. Those were used for EMG data reduction (see below).

Data reduction and EMG threshold determination

Using MATLAB v.2017a, individual contractions were identified using the '*Peakfinder*' function (Yoder 2011) applied to the position data. The EMG root mean square (RMS) of each contraction (EMG_{RMS}) was calculated over a 2-s window where knee position was most stable. Five EMG_{RMS} values were calculated for each load and then averaged ($EMG_{RMS}5$). Bi-segmental regressions of the $EMG_{RMS}5$−%1RM relationships, as determined by the least sum of squares (LSS), were determined (Prism Ver.8; GraphPad Software, San Diego, CA, USA) and used to determine the $EMG_{Th.}$ The latter was defined as the exercise intensity (%1RM) at the intersection of the two resulting regression lines. If 1, 2, or 3 points deviated more than 2, 3, or 4 standard deviations, respectively, from the rest of the regression points they were considered outliers and were removed from the analysis, as long as the newly-determined EMG_{Th} did not deviate by more than ± 1 load from the

original LSS-determined EMG_{Th}, as previously described (Woods et al. 2019).

Statistical analysis

Statistical analysis was performed using SPSS v.24. Data are presented as means ± 1SD. All group data were normally distributed. An independent t-test was used to assess group differences in physical characteristics, and EMG_{Th}-related data between girls and women. A two-way ANOVA was used to investigate age- and sex-related differences in relation to the previous male study (Woods et al. 2019). Results are given as means ± SD. Statistical significance was set at $\alpha < 0.05$.

Results

Representative girl and woman EMG_{RMS} plots are depicted with their respective EMG_{Th} determinations in Fig. 1.

The girls' mean EMG_{Th} occurred at higher relative exercise intensity than the women's ($p < 0.03$, ES = 0.86; Fig. 2). The EMG_{Th}-detection rate was similar in the girls and the women (88.2 vs. 94.1%, respectively). These and other test-performance data are shown in Table 2 and compared with those of the corresponding, earlier male study (Woods et al. 2019). The girls' maximal strength (1RM) was lower than the women's ($p < 0.001$), whether expressed in absolute values or relative to lean body mass (Fig. 3). Test duration (number of completed stages) was longer in the girls, but the difference did not reach statistical significance. The girls' relative exercise intensity at exhaustion (%1RM), however, was significantly higher than the women's ($p < 0.001$).

Table 2 and Fig. 2 also compare the female EMG_{Th} data of the present study with the corresponding male data of our previous study (Woods et al. 2019). The two-way ANOVA analysis for EMG_{Th} indicated a significant main effect of age ($p < 0.01$), but no sex or age-by-sex interactions. That is, EMG_{Th} was higher in children than in adults. Girls' and boys' EMG_{Th} were similar, as were women's and men's corresponding values.

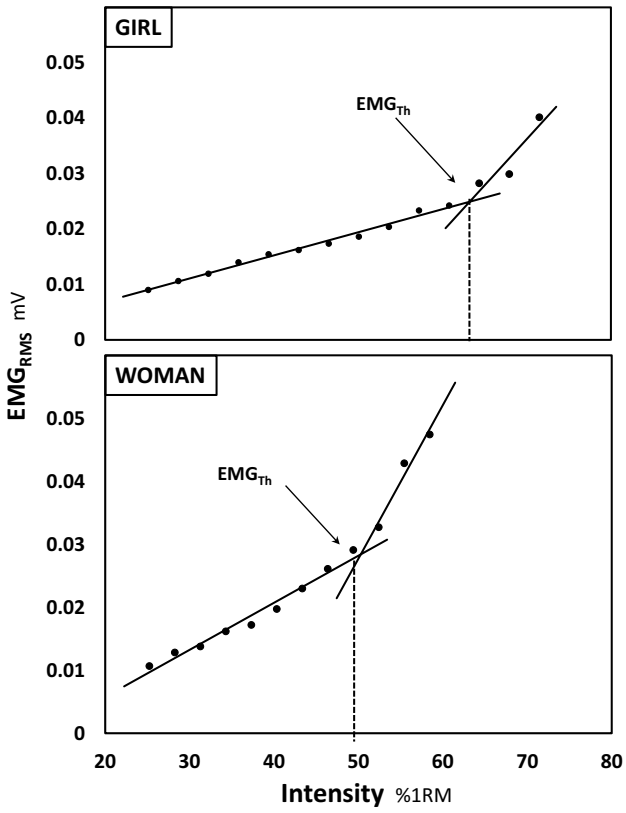

Fig. 1 Examples of a girl's and a woman's EMG_{RMS} plots and their respective EMG_{Th} determinations. Note the difference in EMG_{Th} intensity

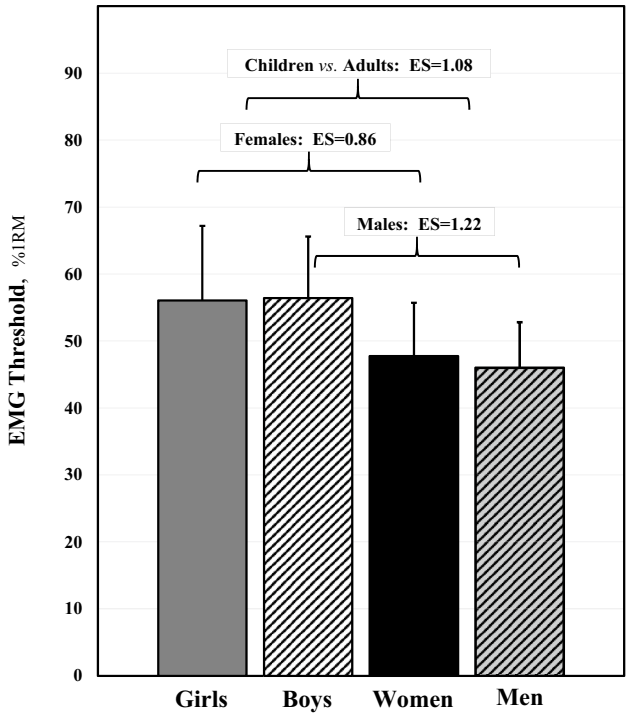

Fig. 2 EMG_{Th} intensity (%1RM) of the girls (56.05) and women (47.72) compared with the boys' (56.4) and men's (46.0) values of our previous male study (Woods et al. 2019). There was a significant age-effect, with high ES for both females (0.86) and males (1.22). Note similarities within age groups

Table 2 Test performance in the current female study vs. previous male study

	Females (present study)		Males (Woods et al. 2019)		Effect, *p*		
	Girls	Women	Boys	Men	Sex	Age	Age × sex
EMG_{Th} detection rate (%)	88.2	94.1	88.9	95.2	NA	NA	NA
1RM, absolute (Nm)	90.4 ± 38.0	222.8 ± 44.3	110.0 ± 23.8	314.5 ± 42.3	< 0.001	< 0.001	< 0.001
Completed stages (#)	15.2 ± 3.1	14.0 ± 1.7	17.2 ± 2.5	15.8 ± 3.3	< 0.005	= 0.05	NS
Load at exhaustion (%1RM)	68.3 ± 7.4	62.9 ± 5.6	76.8 ± 8.5	65.8 ± 9.1	< 0.001	< 0.001	< 0.001

Values are means ± 1SD

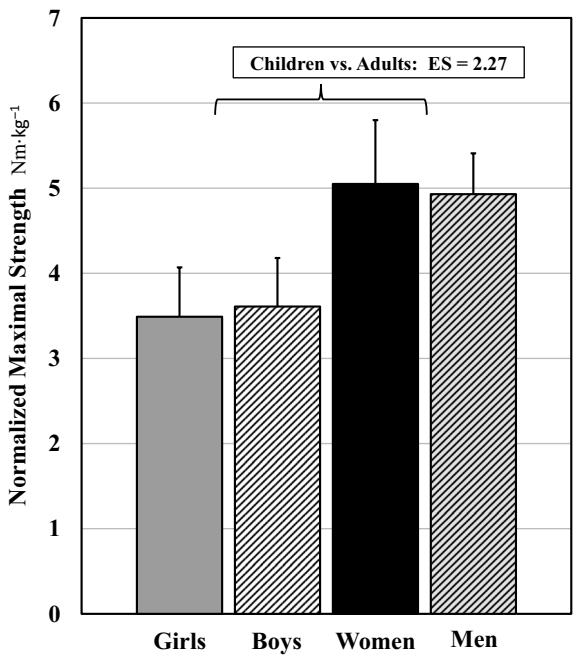

Fig. 3 Comparison of LBM-normalized relative strength of the girls and women of the present study (3.49 and 5.05 Nm kg^{-1}, respectively) with the boys and men of the previous males' study (3.61 and 4.89 Nm kg^{-1}, respectively; Woods et al. 2019)

Discussion

The main finding of the present study was the higher EMG_{Th} values in girls compared with women. This finding extends previous findings in males, and suggest that, when working at similar relative intensities, children recruit their higher-threshold MUs later and to a lesser extent compared with adults. Using the isometric-based test protocol, EMG_{Th}-detection rates among both girls and women were on par with those previously found in males. We expected the girls–women EMG_{Th} difference to be smaller than the corresponding male difference. However, while the observed difference was indeed in the expected direction (8.3 vs. 10.4%), it could not be shown to be statistically significant.

Our finding of higher (or later) EMG_{Th} in the girls extends the previous findings in boys (Woods et al. 2019) and lends further support to the suggestion that, when working at similar relative intensities, children recruit their higher-threshold MUs to a lesser extent compared with adults. Associating children's higher EMG_{Th} with their greater activation deficit is based on known MU recruitment hierarchy (Henneman et al. 1965; Vollestad et al. 1984). The Henneman size principle implies that the unrecruited MUs constituting the activation deficit, are largely or solely of the higher-threshold, type-II variety. Lesser type-II MU activation may also be due to lower type-II muscle composition (Jansson 1996; Lexell et al. 1992). Regardless of which is the main reason, children's lower functional pool of type-II MUs explains why they require higher relative intensities to activate it and manifest an EMG_{Th}.

Using the isometric-based IICT protocol, the EMG_{Th} detection rate was higher than previously observed in females via a cycling protocol (Long et al. 2017) and similar to the corresponding males' rate when using the IICT protocol (Woods et al. 2019) (Table 2). Contrary to men, who had high (95.2%) detection rates under both protocols, the women's detection rate improved 38% relative to cycling (94.1 vs. 68.4%; $\chi^2 = 3.78$, $p = 0.05$). Most dramatic, however, was the 96% detection boost among the girls (88.2 vs. 45.0%; $\chi^2 = 7.54$, $p < 0.01$). This improvement, particularly in the children, appears attributable to the higher, isometrically-attained relative contractile forces achieved in the IICT, compared with those attainable in continuous dynamic cycling (Woods et al. 2019). The higher forces presumably facilitated the detection of higher-intensity thresholds, previously undetectable by the cycling-based protocol. Thus, the IICT appears to be a more sensitive EMG_{Th}-detection tool compared with cycling, manifested by the 49% greater girls−women EMG_{Th} difference compared with cycling (8.33 vs. 5.6%) (Long et al. 2017). This difference is similar to the corresponding ~ 53% increase in the EMG_{Th} difference among the males (10.1 vs. 6.6%) (Woods et al. 2019).

An aspect of the enhanced age-related EMG_{Th} difference is the relative EMG_{Th} percentage ranges of the two exercise modalities (isometric- vs. cycling-based protocols). The girls' cycling-based EMG_{Th} of 88.6% was 6.7% higher

relative to the women's 83.0% [((88.6–83.0)/83.0)*100] (Long et al. 2017). However, the corresponding IICT-based difference of the present study was 17.5% (56.05 vs. 47.72%), or ~ 2.5 times greater than in cycling (~ 2.7 in the males). Importantly, along with our previous findings in males (Woods et al. 2019), this study extends the cycling-based child–adult EMG_{Th} difference observed earlier (Long et al. 2017; Pitt et al. 2015).

The girls' maximal relative strength (LBM-normalized 1RM) was considerably lower than that of the women, which closely corresponded with the previously observed male values (Woods et al. 2019) (Fig. 3). Thus, male and female's relative strength was similar across both age groups with children having only ~ 70% of adult strength. Since strength was LBM-normalized, children's lower strength cannot be attributed to their smaller mass. Neither can the ~ 30% difference be explained away by their ~ 12% lower relative muscle mass (Malina 1969). Most of this difference must therefore be accounted for by children's lower maximal volitional muscle activation (Grosset et al. 2008; O'Brien et al. 2010) and by their suggested lower type-II MU activation, relative to adults (Dotan et al. 2012).

Figure 4 depicts the correlation between relative strength among children and adults [of both the present (female) and previous (male) study (Woods et al. 2019)] and what can be construed as the type-II MU relative contribution to maximal strength (100% − $EMG_{Th,}$ expressed as %1RM). While not proving causality, the correlation explains > 15% of the observed variance and provides further support to the notion that children's lower muscular performance capacity, such as maximal force, is partly due their lower relative ability to recruit their higher-threshold/type-II MU pool, relative to adults.

The observed 8.3% age-related EMG_{Th} difference in females was somewhat smaller than the corresponding 10.4% difference in males (Table 2). Although this difference was not statistically significant (possibly due to insufficient statistical power), it is in line with various studies that reported similar or even larger sex effects in age-related differences in other physiological variables, such as peak and mean power, maximal strength, and rate of torque development (Blimkie 1989; Seger and Thorstensson 1994). A distinct difference between the female and the previous male findings is that while the age-by-sex interaction was not statistically significant, the effect size was larger in males (ES = 1.22) than in females (0.86). A possible explanation for this difference is the fact that the females and particularly the girls, reached lower contractile intensities at test termination than their male counterparts. This was particularly evident in the girls, relative to the boys (Table 2). Notably, 12 of the 17 girls failed to reach 70% 1RM, while all the boys surpassed that intensity (Woods et al. 2019). Since two thresholds can sometimes be detected [e.g., (Hug et al. 2003)], the 2nd of which being typically more acute, low exhaustion intensity might miss the 2nd-threshold altogether and misidentify the 1st as the 2nd threshold (due to the LSS-based detection method). Figure 5 illustrates a two-threshold case that, if terminated at ~ 70% 1RM or lower, would have missed the 2nd threshold. Indeed, the three girls with lowest EMG_{Th} values (< 44% 1RM) reached < 64% 1RM at exhaustion. Omitting those three girls from analysis would result in a higher girls–women effect size (1.56 vs. 0.86) and greater EMG_{Th} difference (12.4 vs. 8.3%). Thus, if this argument is valid for even one or two of the girls then the girl–woman EMG_{Th} difference may be on par or even higher than the corresponding males' value.

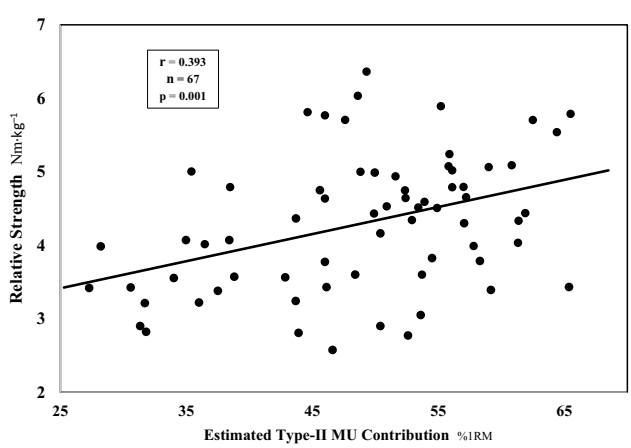

Fig. 4 Relationship between relative strength (LBM-normalized 1RM) and the presumed type-II motor-unit contribution to 1RM force (100 − EMG_{Th}) for the combined participant pool (n = 67) of the present (female) and the previous male study (Woods et al. 2019)

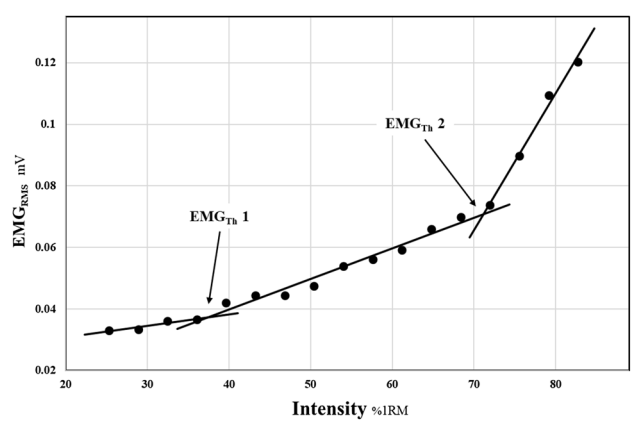

Fig. 5 Example of EMG_{RMS} record of a girl who reached > 80% 1RM at exhaustion. Two thresholds are apparent. Using our LSS-based threshold-detection method, the 2nd (more acute) threshold is identified as the EMG_{Th}. However, had the girl terminated the test at ~ 70% 1RM, or less, only the 1st threshold would have been detected as if it were the 2nd or the only one

The preceding discussion highlights the possibility that while isometric-based testing appears to increase sensitivity and raise EMG_{Th} detection rate relative to cycling-based testing, pre-mature test termination may still occur with the IICT protocol. While both the girls' and women's relative strength was on par with the males, the reason(s) for their lower relative contractile intensity at exhaustion, particularly in the girls (Table 2) is unclear. Whether physiological or psychological, the reasons for this remain to be elucidated in future research. Modifications of IICT's parameters (contraction duration and number per load, recovery times, and load increments) might further improve its overall sensitivity for all.

In the cycling-based protocol, women had only 68% detection rate (Long et al. 2017), but in the IICT they achieved EMG_{Th} detection rate on par with the men's (94 vs. 95%). However, in light of the men−women similarities in both EMG_{Th} and relative strength, it remains to be explained why the women's 68% cycling-based detection rate was so much lower than the men's 95%.

Conclusions

Girls' EMG_{Th} values were higher compared with women's, possibly reflecting their lower ability to activate higher-threshold (type-II) motor units. The larger child−adult differences found using the IICT, compared with the earlier cycling-based protocol, promotes isometric-based testing as a superior means for eliciting higher contractile intensities and, in turn, greater EMG_{Th} detection capacity. As the females' EMG_{Th} and relative strength values were similar to those we previously reported for males, no sex- or maturation-related differences were observed. Children's higher EMG_{Th}, compared with adults, having now been shown in four separate studies, is suggested to reflect their lesser employment of higher-threshold/type-II MUs. This premise, in turn, could prove instrumental for establishing child-optimized physical training and rehabilitation practices that heretofore have largely been based on adult data.

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Authors' contribution SW, RD and BF conceived and designed research. SW, NJ, and RD conducted experiments. RD designed and built the custom-made ergometer. SW and RD analyzed data. SW, RD and BF wrote the manuscript. All authors read and approved the manuscript.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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